Hyperspectral Unmixing via Latent Multiheterogeneous Subspace

Chunzhi Li, Yonggen Gu, Xiaohua Chen, Yuan Zhang, and Lijian Ruan

***Abstract*— Blind hyperspectral unmixing (BHU) is an important technology to decompose the mixed hyperspectral image (HSI), which is actually an ill-posed problem. The ill-posedness of the BHU is deteriorated by nonlinearity, endmember variability (EV) and abnormal points, which are considered as three challenging intractable interferences currently. To sidestep the challenges, we present a novel unmixing model, where a latent multidiscriminative subspace is explored and the inherent self-expressiveness property is employed. The most existing unmixing approaches directly decompose the HSI utilizing original features in an interference corrupted single subspace, unlike them, our model seeks the underlying intrinsic representation and simultaneously reconstructs HSI based on the learned latent subspace. With the help of both clustering homogeneity and intrinsic features selection, structural differences in the HSI and the spectral property of a certain material are exploited perfectly, and an ideal multiheterogeneous subspace is recovered from the heavily contaminated original HSI. Based on the multiheterogeneous subspace, the reconstructed differentiated transition matrix is split into two matrices to avoid the emergence of the artificial endmember. Experiments are conducted on synthetic and four representative real HSI sets, and all the experimental results demonstrate the validity and superiority of our proposed method.**

***Index Terms*— Blind hyperspectral unmixing (BHU), inherent self-expressiveness property, multiheterogeneous subspace, sparse analysis.**

NOMENCLATURE

THROUGHOUT this article, the elementwise product and division are denoted by $.*$ and $./$, $()^{\cdot-(1/2)}$ represents elementwise power of $-(1/2)$, and $Tr(\cdot)$ denotes the trace of a matrix. We use $(\cdot)^T$ to denote the transposition of a variable, to denote the pseudo inverse of a matrix via $(\cdot)^{-1}$, and let $\|\cdot\|_{1/2}$ represent $L_{1/2}$ norm.

BHU	Blind hyperspectral unmixing.
HSI	Hyperspectral data.
EV	Endmember variability.
LMM	Linear mixing model.
RCMF	Robust constrained matrix factorization.
NLMM	Nonlinear mixing model.
BSS	Blind source separation.
NMF	Nonnegative matrix factorization.
KNN	K-nearest neighbors.
SCLSU	Scaled constrained least squares unmixing.
SAD	Spectral angle distance.
RMSE	Root mean square error.
SNR	Signal-to-noise ratio.
$\mathbf{X} \in \Re^{B\times N}$	The matrix of a HSI, where B and N denote the number of spectral channels and pixels in the HSI.
$\mathbf{U} \in \Re^{B\times K}$	The endmember matrix that contains K endmember signatures.
$\mathbf{V} \in \Re^{K\times N}$	The fractional abundance matrix.
$\mathbf{E} \in \Re^{B\times N}$	The reconstructed error matrix, generally, considered as additive noise.
$g_1(\cdot), g_2(\cdot)$	Norm function for sparse constraint.
$\mathbf{P} \in \Re^{B\times B}$	A projection coefficient matrix for pursuing the latent matrix.
$\mathbf{L} \in \Re^{B\times N}$	The exploited latent matrix.
$\mathbf{A} \in \Re^{N\times N}$	A reconstructed transition matrix.
$\mathbf{E}_1 \in \Re^{B\times N}$	The reconstructed error matrix from the process of calculating latent subspace.
$\mathbf{E}_2 \in \Re^{B\times N}$	The reconstructed error matrix from the process of calculating the transition matrix.
$\mathbf{Y} \in \Re^{N\times K}$	An adjacent matrix connects the latent matrix $\mathbf{L}$ and the abundance matrix $\mathbf{V}$.
$\mathbf{W}$, $\widehat{\mathbf{W}}$	Graph weight matrices obtained from the spectral side and spatial side, respectively.
$\mathbf{C}$, $\widehat{\mathbf{C}}$,$\mathbf{D}$, $\widehat{\mathbf{D}}$	The Laplacian matrices correspond to $\mathbf{W}$ and $\widehat{\mathbf{W}}$. Diagonal matrices obtained from $\mathbf{W}$ and $\widehat{\mathbf{W}}$.
Θ, Ξ, Υ, Ψ	The Lagrange multipliers correspond to the variables $\mathbf{P}$, $\mathbf{L}$, $\mathbf{Y}$ and $\mathbf{V}$.
$\mathbf{S}_1 \in \Re^{B\times N}$	A scaled matrix for initializing the projection matrix $\mathbf{P}$.
$\mathbf{S}_2 \in \Re^{1\times N}$	The standard deviation matrix of the HSI matrix.
σ, τ	Kernel size parameters for scaled matrix $\mathbf{S}_1$ and the graph weight matrices.
α, $\alpha_{1\sim5}$	Tradeoff parameters for the unmixing model.

Manuscript received December 31, 2019; revised April 13, 2020; accepted May 18, 2020. Date of publication June 2, 2020; date of current version December 24, 2020. This work was supported in part by the Zhejiang Provincial Science and Technology Key Plan of China under Grant 2017C02036, in part by the National Natural Science Foundation of China under Grant 61501184, in part by the Science and Technology of Zhejiang Province under Grant LY17F020007, in part by the Zhejiang Province Key Laboratory of Smart Management and Application of Modern Agricultural Resources under Grant 2020E10017, in part by the Huzhou Science and Technology Key Plan of China under Grant 2016ZD2011, and in part by the Huzhou Public Welfare Application Research Project under Grant 2019GZ02. *(Corresponding author: Xiaohua Chen.)*

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Digital Object Identifier 10.1109/TGRS.2020.2996249

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β_1, β_2	Tradeoff parameters for initializing the projection matrix $\mathbf{P}$.
$\mathbf{P}_0$, $\mathbf{L}_0$	The initial matrices for $\mathbf{P}$ and $\mathbf{L}$.
$\mathbf{Z} \in \Re^{B \times B}$	A blurring matrix for simulating the nonlinearity.

I. Introduction

HSI is a collection of data set taken across wealthy hundreds of different spectral bands; however, this abundant spectral information comes at a price. The spatial resolution of the HSI is very low, which causes many "mixed pixels" [1]. Actually, HSI plays an important role in many different fields, such as evaluating vegetation condition, estimating environment pollution and urbanization processes [2], but unfortunately, those mixed pixels hinder the application of the HSI. To improve the performances of the HSI in applications, the BHU becomes an indispensable technology. It can simultaneously decompose the HSI into "pure single materials" and corresponding fractional components, known as "endmembers" and "abundances" [1], [2], respectively.

A lot of work has been invested in the BHU. Since LMM and NLMM are capable of well modeling the real mixing process, many existing unmixing approaches are based on the two canonical models. To address the challenging problems in the BHU, the state-of-the-art unmixing methods can be mainly classified into three series, that is, focusing on tackling the nonlinearity, considering EV, and advocating robust unmixing methods to mitigate the influence of the outlier. In terms of nonlinearity, nonlinear methods play a key role in improving the performance of unmixing. In nonlinear unmixing methods, intimate mixtures of minerals and multiple scatterings can be tackled by bilinear mixture models [3], multilinear mixing models [4], kernel-based nonlinear models [5], extended multilinear mixing models [6], graphical models [7], and so on. Due to varying imaging conditions, the spectral substances contained in an observed HSI can change spectrally, spatially, or temporally from an HSI to another [8], [9], which is the issue of EV. It can cause serious estimation errors and has received an increasing attention. Some related works, such as, to tackle the EV by employing adaptive bundles and double sparsity [10]; considering the EV by a perturbed linear mixing model [8]; literature [11] treats EV via two measures: using the extended linear mixing model [12] to tackle the EV induced by illumination, and dealing with intrinsic variability effects by using a statistical approach [11]. Accounting for EV and abrupt spectral changes by a hierarchical Bayesian model [13], and so on. As for the outlier, also called the abnormal point, it can easily lead to large increase in construction error [14], to this end, a great number of research studies are devoted to develop robust algorithms, such as an LMM-based Bayesian method [15], an outlier-insensitive unmixing method with spatially varying hyperspectral signatures [16], RCMF for the BHU [17], a robust unmixing algorithm via correntropy-induced metric [14]. Most of the above-mentioned unmixing methods reconstruct the data points on the original single subspace directly, but the others employ the single space by subtracting the reconstruction error and generate specific representations with different constraints. Therefore, the concerned interferences are tackled simultaneously or separately.

Based on the above analysis, we can conclude that the original features can dramatically influence the performance of the unmixing methods, particularly when the HSIs are grossly corrupted, as a result, helpful data points are scanty as covered by the interferences and/or noises, and what is worse, the artificial endmember[1] would emerge. Motivated by the existing unmixing algorithms, we concern several interesting questions.

1) The HSI contains the interferences from EV, nonlinearity and outlier, these interferences may vary spectrally and spatially, interference/noise—which is omnipresent—further and raises the difficulty in unmixing on the original feature space. How to effectually deal with the noise and interference and improve the quality of the feature space?
2) Due to the complicated physical imaging mechanism, the HSIs possess heterogeneous features collected from diverse imaging conditions, how to extract the endmember signatures and abundances from multiheterogeneous features?
3) The multiheterogeneous features in the HSI hold rich information from multiple insights, which could be helpful to unmixing. However, the present BHU approaches usually makes the unmixing by riskily using only one full-view subspace. The single undifferentiated subspace alone is generally not sufficient to represent data points, how to exploit the relationships among data points?
4) Considering the ill-posedness of the BHU problem, data points vary from the plurality have great influence on the unmixing results, that is, one outlier can destroy the archetype solution and bring about artificial endmembers.[1] How to avoid the emergence of the meaningless signature or artificial endmember?

Since the imaging of the HSI is interfered by nonlinearity, EV, and outliers, and the distribution of these interferences usually are heterogeneous, in addition, different endmembers locate at different information subspaces, thus, multiheterogeneous information emerges in the HSI, that is, various modalities are generated for the BHU task. Moreover, due to the ill-posedness of the BHU, the artificial endmember usually emerges in the solution. Turning attention to the theory of subspace clustering, we can find that multiview subspace clustering approach is a powerful tool to cluster [18], [19], which can exploit the complementary information from multiple subspaces and describe each data point [20], [21]. Inspired by the subspace clustering and motivated by avoiding emerging the artificial endmember, we focus on addressing the above challenging problems by exploiting latent multiheterogeneous subspace and employing the inherent self-expressiveness property. The proposed method exploits an intrinsic underlying space via manifold sparse learning to overcome the diverse inherent interferences of the HSIs and maximizes the mutual agreement

[1]It is not a real substance in the scene. Due to the ill-posedness of BHU problem, artificial endmember just exists in the solution space, which is meaningless from the physical perspective.

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of the different endmember signatures with the help of the correlations from the different data points. Moreover, by means of the subspace clustering theory, the faithful similarity structures of the HSI can be preserved in the multiple discriminative subspaces. Therefore, the proposed method can maximize the interclass distance and concurrently minimize the intraclass distance. In particular, the accuracy of the endmember signature is guaranteed by the inherent self-expressiveness property. The main contributions of our proposed method are summarized as follows.

1) We treat different endmember signatures and perturbations as multiheterogeneous structural information, and multiview subspace clustering theory is exploited to pursue latent multiheterogeneous subspaces. A noise/interference[2]-insensitive latent space is exploited by manifold sparse learning and an additional sparsity constraint.
2) Instead of extracting the endmembers and corresponding abundances with the original HSI directly. The exploited latent space can explore the clustering homogeneity and capture intrinsic features, and the BHU task can obtain multiple information from the multiheterogeneous with the support of data-dependent information, consequently, the interferences are mitigated and the underlying intrinsic features in the HSI are fully exploited.
3) The proposed unmixing model possesses the inherent self-expressiveness property, which can mitigate the emergence of the artificial endmember.
4) The significance of the underlying space is validated experimentally. Moreover, to demonstrate the effectiveness of exploiting the underlying discriminative subspace and the robustness of tackling the artificial endmember, extensive experiments are implemented on synthetic and four broadly used real data sets. In all cases, the proposed method performs better than the compared methods. Furthermore, each constraint is validated to be helpful to the unmixing performance.

The remainder of this article is organized as follows. Section II states the unmixing problem and gives a traditional unmixing algorithm. In Section III, a novel unmixing model is proposed, then a corresponding algorithm is introduced, especially, an initializing algorithm for the projection matrix is given in this section. In Section IV, the employed data sets, compared methods, superiority of the underlying subspace and evaluated metrics are illustrated. A series of experimental results and analyses are illustrated to demonstrate the performance of the proposed method in Section V. Finally, we summarize this article in Section VI.

II. Related Works

Generally, an LMM-based BHU problem can be depicted as

$$\mathbf{X} = \mathbf{UV} + \mathbf{E} \tag{1}$$

where $\mathbf{X} \in \Re^{B\times N}$ denotes the HSI matrix with the number of N pixels across B spectral bands, $\mathbf{U} \in \Re^{B\times K}$ denotes the endmember matrix and contains the number of endmember signatures as K, $\mathbf{V} \in \Re^{K\times N}$ is the corresponding abundances, and $\mathbf{E} \in \Re^{B\times N}$ is the reconstruction error matrix, it is commonly considered as the additive noise [14], [17]. According to the physical interpretations, the matrices of $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{E}$ should be nonnegative, and the abundance matrix should satisfy the condition of sum-to-one constraint.

Note that the endmember matrix and the abundance matrix are both unknown; thus, BHU problem is similar to the problem of BSS, and a well-known tool called NMF for BSS is usually employed to solve the BHU problem. Here, we take a typical NMF-based algorithm to illustrate the solving process for (1). Under the NMF framework and the physical constraints, (1) can be written as

$$\begin{aligned} &\min_{\mathbf{U},\mathbf{V}} \ \frac{1}{2}\|\mathbf{X}-\mathbf{UV}\|_F^2 + \zeta g_1(\mathbf{V}) \\ &\text{s.t. } u_{bk} \geq 0, \quad v_{kn} \geq 0, \ \sum_{k=1}^{K} v_{kn} = 1 \end{aligned} \tag{2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $g_1(\mathbf{V})$ is a sparsity constrained function for abundance matrix $\mathbf{V}$, such as sparse norm function [22], manifold regularized sparse constraint [23], and substance dependence constrained sparse term [24]. And ζ is a tradeoff parameter. Let $g_1(\mathbf{V})$ be a conventional norm function $g_1(\mathbf{V}) = \|\mathbf{V}\|_{1/2}$, written as

$$\|\mathbf{V}\|_{\frac{1}{2}} = \sum_{k=1}^{K}\sum_{n=1}^{N} v_{kn}^{\frac{1}{2}}. \tag{3}$$

Then the loss function in (2) can be iterated by the following formulas, which have been presented in [25]:

$$\begin{aligned} \mathbf{U} &\leftarrow \mathbf{U}.*\left(\mathbf{XV}^T./\mathbf{UVV}^T\right) \\ \mathbf{V} &\leftarrow \mathbf{V}.*\left(\mathbf{U}^T\mathbf{X}./\left(\mathbf{U}^T\mathbf{UV}+\frac{1}{2}\zeta\mathbf{V}^{\cdot^{-\frac{1}{2}}}\right)\right) \end{aligned} \tag{4}$$

where $.*$, $./$, and $^{\cdot^{-(1/2)}}$ denote elementwise multiplication, division, and $-(1/2)$ power, and $(\cdot)^T$ denotes the calculation of matrix transposition.

It is obvious that the gross matrix $\mathbf{X}$ is treated as a full-view space and utilized to unmix directly. Although the sparsity for abundance matrix $\mathbf{V}$ is considered by the term $g_1(\mathbf{V})$, promising results have been obtained, however, the interferences and noises in the HSI can propagate throughout the iterations to deteriorate the unmixing results. Moreover, considering some useful data points are covered by interferences and/or noises, multiple cues should be exploited to unmix, and worse yet, the correlations among the data points are unavailable in (4). Additionally, conventional unmixing algorithms usually neglect to constrain the iterating process for the endmembers, as a result, some artificial endmembers may be obtained for the ill-posedness of the BHU problem, which are not real existent substances and senseless.

III. Problem Formulation

In recent years, the interferences of EV, nonlinearity, and outlier have attracted more and more attention, and to some extent, the ill-posedness of the BHU is improved by tackling

[2]Throughout this article, interferences mean that the disturbing factors are from EV, nonlinearity, and outlier, while noise means the conventional additive noise.

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these interferences. However, the multiple interferences/noise and the different signatures lead to multiheterogeneous distribution of the HSI, the correlation among multiheterogeneous modalities is usually neglected. In this article, we treat the interferences from an innovative perspective to develop a novel particular method for solving the challenging issues in the BHU.

A. Latent Multiheterogeneous Subspace Analysis

In fact, the data points for a material signature in the HSI are related in some ways; and the data points are together with the abovementioned interferences, which are associated with the endmember signature directly or implicitly, thence giving different modalities for the BHU task. To avoid the propagation of the interferences across the unmixing process, we presume that there exists a latent feature space for those different endmembers, which contains information from several different views and provides us a more comprehensive perspective of the BHU task, thus a latent space is exploited to substitute for the original matrix $\mathbf{X}$. In particular, considering the interferences brings about multiheterogeneous structure and the different endmembers locate at different subspaces, the theory of subspace clustering is employed for analyzing the different modalities in the BHU task, and a transition discriminative matrix $\mathbf{A}$ is introduced, described as

$$\begin{aligned}\min_{\mathbf{P},\mathbf{L},\mathbf{A}} \ & \frac{1}{2}\|\mathbf{X}-\mathbf{PL}\|_F^2+\frac{1}{2}\|\mathbf{L}-\mathbf{LA}\|_F^2+\frac{1}{2}\alpha_1 Tr\left(\mathbf{P}^T\mathbf{CP}\right)\\ & \alpha_2\|\mathbf{P}\|_{\frac{1}{2}}+\beta g_2(\mathbf{A})\\ \text{s.t. } & p_{bb}\geq 0,\quad l_{bn}\geq 0,\ a_{nn}\geq 0\\ & \mathbf{E}_1=\mathbf{X}-\mathbf{PL};\quad \mathbf{E}_2=\mathbf{L}-\mathbf{LA}\end{aligned} \tag{5}$$

where $\mathbf{P}\in\Re^{B\times B}$ is a heterogeneous coefficient matrix, which predicts endmember identification from the point in the underlying feature space $\mathbf{L}$, and $\mathbf{L}\in\Re^{B\times N}$ denotes the latent space, which is expected to be an intrinsic multidiscriminative subspace without any interference or noise. $\mathbf{A}\in\Re^{N\times N}$ is a reconstructed transition matrix for the BHU problem, which is a linear representation such that each modality is denoted by a linear combination of pixels belonging to the same subspace, that is, the matrix $\mathbf{A}$ contains the information of the relationships among the pixels, considering the sparsity of the discrepancy component, the sparsity of the transition matrix should be guaranteed, a sparse function $g_2(\mathbf{A})$ is imposed on the matrix $\mathbf{A}$. α_1, α_2, and β are tradeoff parameters. $Tr(\cdot)$ denotes the trace of a matrix. $\mathbf{E}_1\in\Re^{B\times N}$ and $\mathbf{E}_2\in\Re^{B\times N}$ are, respectively, the reconstructed errors from the processes of obtaining latent subspace and the transition matrix. In particular, to limit coefficient values of the projection matrix $\mathbf{P}$, and guarantee the intrinsic features can be fully exploited, $\mathbf{P}$ is enforced by a sparse constraint, that is, $\|\mathbf{P}\|_{\frac{1}{2}}$. Moreover, manifold learning is employed to pursue the intrinsic manifold structure from the spectral side, and according to the heat kernel [23], the graph weight matrix is calculated

$$\mathbf{W}_{ij}=e^{-\frac{\|x_i-x_j\|^2}{\tau}}. \tag{6}$$

Here, x_i is a KNN of x_j in the spectral space, τ is kernel size parameter, then Laplacian matrix $\mathbf{C}\in\Re^{B\times B}$ in (5) can be deduced as $\mathbf{C}=\mathbf{D}-\mathbf{W}$, and $\mathbf{D}$ is a diagonal matrix, obtained by $\mathbf{D}_{ii}=\sum_j\mathbf{W}_{ij}$.

The superiorities and related justifications of (5) are stated as follows.

1) Manifold learning is employed to pursue a nonlinearity-insensitive space. Moreover, since the intrinsic endmember features distribute in sparse space, sparsity constraint is imposed on the label projection matrix $\mathbf{P}$. Based on the two constraints, an ideal projection matrix $\mathbf{P}$ can be captured to pursue a high-quality latent matrix $\mathbf{L}$, which contains intrinsic features but are intensive to the superficial and changeable interferences and noise.
2) Actually, each pixel in the obtained latent matrix $\mathbf{L}\in\Re^{B\times N}$ belongs to a union of K endmember signatures. Each endmember signature contains n_i pixels with $\sum_{i=1}^{K}n_i=N$. Let $\mathbf{L}_i\in\Re^{B\times n_i}, i=1,2,\ldots,K$ represent the subimage in $\mathbf{L}$ that belong to endmember i. With the help of the heterogeneous subspace structure, the interference/noise-free data points in the latent matrix $\mathbf{L}$ follow the so-called inherent self-expressiveness property, that is, each data point in a set of endmember subspaces can be well denoted by the following expression:
$$\mathbf{L}\approx\mathbf{LA}. \tag{7}$$
The model (7) is famous for the performance in subspace clustering [26], [27]. The goal of selection matrix $\mathbf{A}$ is to obtain certain $\mathbf{A}$ such that it is differentiated for subspace clustering. Here, we aim to look for a linear transition representation matrix $\mathbf{A}$, so then each endmember is represented as a linear combination of signatures belonging to the same subspace, that is, $\mathbf{L}_i=\mathbf{L}_i\mathbf{A}_i$, $i=1,2,\ldots,K$, therefore, matrix $\mathbf{A}$ is expected to be a sparse matrix.
3) Since there exist some correlations between the data points such as the material in a HSI, that is, the spatial similarities in the same material. Hence, it can be concluded that the spatial structure information is a crucial cue to maintain faithful feature in the unmixing process. From another point of view, matrix $\mathbf{A}$ can be considered as a spatial similar distance matrix, which reveals the relationship among pixels by the adaptively assigned distances of the neighbors in a similar structure, consequently, the unmixing can preserve the iteratively reconstructed collaborative similarity structure.
4) In practical application, the dimensions of the intrinsic feature subspaces are mainly determined by the capabilities of the model (7) on maintaining the similarity structures in the multiple discriminative subspaces [28]. In light of the above analyses, (5) can adequately exploit the discriminative subspace structure and the correlations between the data points via the theory of model (7). Consequently, the information covered by the interferences of EV, nonlinearity, and outlier can be exploited via pursuing a robust latent feature matrix $\mathbf{L}$ and constructing discriminative subspace, in other words, the interferences of EV, nonlinearity, outlier can be tackled effectively.

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B. Unmixing Model

To calculate the endmember and abundance matrices, the differentiated transition matrix $\mathbf{A}$ is split into coefficient matrix $\mathbf{Y}$ and abundance matrix $\mathbf{V}$. Based on (5), matrix $\mathbf{A}$ is captured for clustering, and the task of the BHU benefits from the merits of the underlying feature space $\mathbf{L}$ and the model (7). However, the emerging virtual endmembers in the solution space should be considered. Fortunately, the term $\mathbf{L}-\mathbf{LA}$ in (5) possesses the inherent self-expressiveness property [26], [27], which is helpful in avoiding the emergence of the artificial endmember. Therefore, endmember matrix is calculated as $\mathbf{U}=\mathbf{LY}$,[3] that is, (5) is transformed into the following equation:

$$\mathbf{L} \approx \mathbf{LYV} \tag{8}$$

where $\mathbf{Y} \in \Re^{N\times K}$ is an unknown nonnegative adjacent matrix, which records the relationship between mixtures and the pure pixels. Particularly, the matrices $\mathbf{Y}$ and $\mathbf{V}$ are invariant to the scaling factor of the data and the affine transformation, it means that the matrices $\mathbf{Y}$ and $\mathbf{V}$ are immune to the scale-factors. According to the analysis on the model (7), matrix $\mathbf{A}$ should satisfy the condition of sparsity, thus $\mathbf{Y}$ and $\mathbf{V}$ should be sparse. To guarantee the sparsity of matrices Y and V, sparsity constraints are imposed on the matrices Y and V. In doing so, $\mathbf{YV}$ is controlled to be as sparse as matrix $\mathbf{A}$. Finally, the unmixing model is formulated as

$$\begin{aligned}\min_{\mathbf{P},\mathbf{L},\mathbf{U},\mathbf{V}} & \frac{1}{2}\|\mathbf{X}-\mathbf{PL}\|_F^2+\frac{1}{2}\|\mathbf{L}-\mathbf{LYV}\|_F^2 \\ & +\alpha\left[\frac{1}{2}Tr\left(\mathbf{P}^T\mathbf{CP}\right)+\|\mathbf{P}\|_{\frac{1}{2}}+\|\mathbf{Y}\|_{\frac{1}{2}}\right. \\ & \left.+\|\mathbf{V}\|_{\frac{1}{2}}+\frac{1}{2}Tr\left(\mathbf{V}\widehat{\mathbf{C}}\mathbf{V}^T\right)\right] \\ \text{s.t. } & p_{bb}\geq 0,\quad l_{bn}\geq 0,\ y_{nk}\geq 0,\ v_{kn}\geq 0,\ U=LY \\ & \mathbf{E}_1=\mathbf{X}-\mathbf{PL},\quad \mathbf{E}_2=\mathbf{L}-\mathbf{LYV}, \\ & \sum_{k=1}^{K}y_{nk}=1,\quad \sum_{k=1}^{K}v_{kn}=1\end{aligned} \tag{9}$$

where $\widehat{\mathbf{C}} \in \Re^{N\times N}$ is a Laplacian matrix that is calculated on the latent matrix space $\mathbf{L}$ from the spatial side, and the corresponding weight matrix is calculated as

$$\widehat{\mathbf{W}}_{ij}=e^{-\frac{\|l_i-l_j\|^2}{\tau}} \tag{10}$$

where l_i and l_j denote the points from the spatial side. Accordingly, $\widehat{\mathbf{C}}=\widehat{\mathbf{D}}-\widehat{\mathbf{W}}$, and $\widehat{\mathbf{D}}_{ii}=\sum_j \widehat{\mathbf{W}}_{ij}$. Although each constraint possesses a unique role in (9), to simplify analysis, the influence of each constraint is considered equivalent, and the tradeoff parameter of all the constraints is set as α. Additionally, sum-to-one constraint is imposed on the abundance matrix, to ensure the implementation of the sum-to-one constraint, the row of the adjacent matrix $\mathbf{Y}$ is also constrained by the sum-to-one constraint.

In contrast to the existing unmixing methods, an inherent latent matrix is pursued by (9) to displace the gross HSI. To achieve the BHU task, the advocated unmixing model is formulated as model (9), consequently, the reconstruction error matrix $\mathbf{E}$ in (1) is split by model (9) into $\mathbf{E}_1$ and $\mathbf{E}_2$. By these and with several sparse constraints from multiple angles, intrinsic stable sparser features are exploited, and multiple discriminative information and the correlations between data points can work together, and as a result, the correctness of the unmixing is guaranteed.

[3]In this article, we just consider the relationship between matrix $\mathbf{L}$ and matrix $\mathbf{Y}$ is linear, and we will extend this relationship in our future works.

C. Iterating Algorithm

It is obvious that the proposed approach (9) is not a convex problem, optimizing $\mathbf{P}$, $\mathbf{L}$, $\mathbf{U}$, and $\mathbf{V}$ together is a NP-hard problem. To overcome this difficulty, the variables are minimized alternately, that is, when one variable is iterating, fixing the other three variables as constants, and each variable is renewed conditionally upon the current fixed variables, in such a way the problem of calculating the proposed model (9) is split into four convex problems, and the convergent proof for each problem can be found in the existing literature [24], [25].

According to the physical interpretations of the variables $\mathbf{P}$, $\mathbf{L}$, $\mathbf{Y}$, and $\mathbf{V}$, all of them are constrained to be nonnegative. Given the nonnegative matrices $\Theta \in \Re^{B\times B}$, $\Xi \in \Re^{B\times N}$, $\Upsilon \in \Re^{N\times K}$, and $\Psi \in \Re^{K\times N}$ as the Lagrange multipliers that correspond to the variables $\mathbf{P}$, $\mathbf{L}$, $\mathbf{Y}$, and $\mathbf{V}$. Define the Lagrange function Γ as

$$\begin{aligned}\Gamma = & \min_{\mathbf{P},\mathbf{L},\mathbf{U},\mathbf{V}}\frac{1}{2}\|\mathbf{X}-\mathbf{PL}\|_F^2+\frac{1}{2}\|\mathbf{L}-\mathbf{LYV}\|_F^2 \\ & +\alpha\left[\frac{1}{2}Tr\left(\mathbf{P}^T\mathbf{CP}\right)+\|\mathbf{P}\|_{\frac{1}{2}}+\|\mathbf{Y}\|_{\frac{1}{2}}+\|\mathbf{V}\|_{\frac{1}{2}}\right. \\ & \left.+\frac{1}{2}Tr\left(\mathbf{V}\widehat{\mathbf{C}}\mathbf{V}^T\right)\right]+Tr\left(\Theta\mathbf{P}^T\right) \\ & +Tr\left(\Xi\mathbf{L}^T\right)+Tr\left(\Upsilon\mathbf{Y}^T\right)+Tr\left(\Psi\mathbf{V}^T\right).\end{aligned} \tag{11}$$

To achieve the task of minimizing the variables alternately, the partial derivative for each variable can be calculated as

$$\frac{\partial\Gamma}{\partial\mathbf{P}}=-\mathbf{XL}^T-\alpha\mathbf{PW}+\mathbf{PLL}^T+\alpha\mathbf{P}^{\cdot^{-\frac{1}{2}}}+\alpha\mathbf{PD}+\Theta; \tag{12}$$

$$\frac{\partial\Gamma}{\partial\mathbf{L}}=-\mathbf{P}^T\mathbf{X}-2\mathbf{LYV}+\mathbf{L}+\mathbf{P}^T\mathbf{PL}+\mathbf{LYVV}^T\mathbf{Y}^T+\Xi; \tag{13}$$

$$\frac{\partial\Gamma}{\partial\mathbf{Y}}=-\mathbf{L}^T\left(\mathbf{LV}^T\right)+\mathbf{L}^T\left[(\mathbf{LY})\left(\mathbf{VV}^T\right)\right]+\alpha\mathbf{Y}^{\cdot^{-\frac{1}{2}}}+\Upsilon; \tag{14}$$

$$\begin{aligned}\frac{\partial\Gamma}{\partial\mathbf{V}}= & -\mathbf{Y}^T\mathbf{L}^T\mathbf{L}-\alpha\mathbf{V}\widehat{\mathbf{W}}+\alpha\mathbf{V}^{\cdot^{-\frac{1}{2}}} \\ & +\alpha\mathbf{V}\widehat{\mathbf{D}}+\mathbf{Y}^T\mathbf{L}^T\mathbf{LYV}+\Psi.\end{aligned} \tag{15}$$

According to the theory of Karush–Kuhn–Tucker, $\Theta.*\mathbf{P}=0$, $\Xi.*\mathbf{L}=0$, $\Upsilon.*\mathbf{Y}=0$, $\Psi.*\mathbf{V}=0$, and the following expressions obey the following conditions:

$$\begin{aligned}&\left(-\mathbf{XL}^T-\alpha\mathbf{PW}+\mathbf{PLL}^T+\alpha\mathbf{P}^{\cdot^{-\frac{1}{2}}}+\alpha\mathbf{PD}\right).*\mathbf{P}=0; \\ &\left(-\mathbf{P}^T\mathbf{X}-2\mathbf{LYV}+\mathbf{L}+\mathbf{P}^T\mathbf{PL}+\mathbf{LYVV}^T\mathbf{Y}^T\right).*\mathbf{L}=0; \\ &\left[-\mathbf{L}^T\left(\mathbf{LV}^T\right)+\mathbf{L}^T(\mathbf{LY})\left(\mathbf{VV}^T\right)+\alpha\mathbf{Y}^{\cdot^{-\frac{1}{2}}}\right].*\mathbf{Y}=0; \\ &\left[-\mathbf{Y}^T\mathbf{L}^T\mathbf{L}-\alpha\mathbf{V}\widehat{\mathbf{W}}+\alpha\mathbf{V}^{\cdot^{-\frac{1}{2}}}+\alpha\mathbf{V}\widehat{\mathbf{D}}+\mathbf{Y}^T\mathbf{L}^T\mathbf{LYV}\right].*\mathbf{V}=0.\end{aligned}$$

Then the updating rule for each variable is deduced as follows:

$$\mathbf{P}\leftarrow\mathbf{P}.*\left[\left(\mathbf{XL}^T+\alpha\mathbf{PW}\right)./\left(\mathbf{PLL}^T+\alpha\mathbf{P}^{\cdot^{-\frac{1}{2}}}+\alpha\mathbf{PD}\right)\right]; \tag{16}$$

$$\mathbf{L}\leftarrow\mathbf{L}.*\left[\left(\mathbf{P}^T\mathbf{X}+2\mathbf{LYV}\right)./\left(\mathbf{L}+\mathbf{P}^T\mathbf{PL}+\mathbf{LYVV}^T\mathbf{Y}^T\right)\right]; \tag{17}$$

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$$\mathbf{Y} \leftarrow \mathbf{Y}.*\left[\left(\mathbf{L}^T\left(\mathbf{L}\mathbf{V}^T\right)\right)./\left(\mathbf{L}^T(\mathbf{L}\mathbf{Y})\left(\mathbf{V}\mathbf{V}^T\right)+\alpha\mathbf{Y}^{\cdot-\frac{1}{2}}\right)\right]; \quad (18)$$

$$\mathbf{V} \leftarrow \mathbf{V}.*\left[\left(\mathbf{Y}^T\mathbf{L}^T\mathbf{L}+\alpha\mathbf{V}\widehat{\mathbf{W}}\right) ./\left(\alpha\mathbf{V}^{\cdot-\frac{1}{2}}+\alpha\mathbf{V}\widehat{\mathbf{D}}+\mathbf{Y}^T\mathbf{L}^T\mathbf{L}\mathbf{Y}\mathbf{V}\right)\right]. \quad (19)$$

Algorithm 1: Procedure for the Proposed Method

Step 1: Input HSI matrix $\mathbf{X}$. Initialize $\mathbf{P}$, $\mathbf{L}$, $\mathbf{Y}$, $\mathbf{V}$, set the number of endmember signatures K, set the number of KNN and the kernel size τ for calculating the manifold graph, and set parameter α.

Step 2: Calculate the graph weight matrix $\mathbf{W}$ on the matrix $\mathbf{X}$ with (6) and calculate $\widehat{W}$ on the matrix L with (10).

Step 3: Repeat

while *stopping condition is not satisfied* **do**

Update $\mathbf{P}$ with (16).

Update $\mathbf{L}$ with (17).

Update the graph weight matrix $\widehat{\mathbf{W}}$ with (10).

Update $\mathbf{Y}$ with (18), and rescale each row of $\mathbf{Y}$ to unit norm.

Update $\mathbf{V}$ with (19), and rescale each column of $\mathbf{V}$ to unit norm.

end

Step 7: Calculate the endmember matrix by $\mathbf{U} = \mathbf{L}\mathbf{Y}$.

Step 4: Output $\mathbf{L}$, $\mathbf{U}$ and $\mathbf{V}$.

1) Instructions for Overall Method: The overall procedure for the advocated unmixing model (9) is illustrated in Algorithm 1, and some essentials are elaborated as follows.

1) In this article, we are keen in developing a BHU method, while estimating the number of endmembers is another research field, the number of endmembers K is obtained by referring to the existing literatures, such as [29].
2) To calculate the manifold graph weight matrices from the spectral and spatial sides, generally, the number of KNN is set to $5 \sim 10$, and the kernel size τ is set in the range of $0 \sim 1$. Moreover, the tradeoff parameter α is a small constant to control the weight of the corresponding constraint. Normally, the value depends on the quality of the input data, and we set them in the range of $1e-3 \sim 1e-10$.
3) With respect to the initial values of $\mathbf{P}$, $\mathbf{L}$, $\mathbf{Y}$, and $\mathbf{V}$, a special initializing algorithm is advocated for $\mathbf{P}$, which will be introduced in detail in the successive part. On the basis of the initialized matrix $\mathbf{P}$, $\mathbf{L}$ is initialized as $\mathbf{P}^{-1}\mathbf{X}$ or let initial $\mathbf{L}$ be equal to $\mathbf{X}$, here, $(\cdot)^{-1}$ denotes the calculation of the pseudo inverse matrix. And the initial endmember matrix $\mathbf{U}$ is obtained by the way of random initialization. In the way of random initialization, the initial components of U are assigned between 0 and 1 randomly. Then the method of SCLSU [30] is employed to initialize the abundance matrix $\mathbf{V}$, and $\mathbf{Y}$ is initialized via the way of random initialization or let $\mathbf{Y}$ be $(\mathbf{V}^T/2)$.
4) It is worth noting that the row of the adjacent matrix $\mathbf{Y}$ should be rescaled to unit norm, which corresponds to the sum-to-one constraint for the abundance matrix, moreover, the sum-to-one constraint is also imposed on the abundance matrix in the iteration process.
5) The stopping conditions are designed as follows, if the norms of the error matrices $\mathbf{E}_1$ and $\mathbf{E}_2$ are both not greater than the given tolerant thresholds successively for ten times, the iteration stops; otherwise, repeating the iterating processes until reaching the maximum number of iterations. Empirically, the tolerances are set to $1e-6$ and the maximum number of iterations is not bigger than 10 000.

2) Initializing Algorithm for $\mathbf{P}$*:* The selection of matrix $\mathbf{P}$ is not unique and problem-dependent. Since the target of coefficient matrix $\mathbf{P}$ is mapping the gross matrix to a latent space, which is a determinative variable to the accuracy of the subsequent unmixing. The initializations are crucial to the ill-posedness of the BHU problem, the interferences of EV, nonlinearity, outlier, and additive noise should be considered, and a robust initializing method for the coefficient matrix $\mathbf{P}$ is proposed to pursue an underlying interference/noise-free space. Concretely, it is designed to possess several abilities, that is, to mitigate the influence of nonlinearity, outliers, EV, and obtain sparse discriminative initialization. Since manifold learning is helpful in tackling nonlinearity and pursue sparsity [23], and the theory of subspace clustering is specialized in capturing discriminative features, additionally, half-quadratic minimization technique [14], [31] can mitigate the influence of outliers, therefore, all of these techniques are employed to initialize matrix $\mathbf{P}$. Together with the constant term, the overall initializing algorithm contains four parts, and the final initializing algorithm is formulated as

$$\mathbf{P}_0 = \frac{\beta_1\mathbf{S}_1.*\mathbf{L}_0\mathbf{L}_0^T+\beta_1\mathbf{L}_0.*\mathbf{S}_2\mathbf{L}_0^T+\alpha\mathbf{W}}{\beta_2\mathbf{L}_0\mathbf{L}_0^T+0.04\mathbf{O}+\beta_1\mathbf{L}_0.*\mathbf{S}_2\mathbf{L}_0^T+\alpha\mathbf{D}} \quad (20)$$

here, $\mathbf{O} \in \Re^{B\times B}$ is an identity matrix to avoid the components of the initial matrix $\mathbf{P}$ becoming too large. $\mathbf{P}_0$ and $\mathbf{L}_0$ are the initial matrices for the matrices $\mathbf{P}$ and $\mathbf{L}$. β_1 and β_2 are tradeoff parameters. $\mathbf{S}_1 \in \Re^{B\times N}$ and $\mathbf{S}_2 \in \Re^{1\times N}$ are deduced by the purpose of mitigating the interferences and noise, and the entries of $\mathbf{S}_2$ are the standard deviations of the HSI matrix $\mathbf{X}$, calculated by

$$\mathbf{S}_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[x_{\star i}-\left(\sum_{i=1}^{N}x_{\star i}\right)/N\right]^2}. \quad (21)$$

$x_{\star i}$ denotes the column vectors, which are from the matrix $\mathbf{X}$. And $\mathbf{S}_1$ is a scaled matrix, defined as

$$s_{1(i)} = \exp\left(-\frac{\| x_{\star i} \|^2}{2\sigma^2}\right);$$
$$\sigma^2 = \frac{1}{2BN}\sum_{i=1}^{N}\sum_{j=1}^{B}\left(x_{ij}\right)^2. \quad (22)$$

exp denotes the exponential function. Except the constant term, the other three parts of the proposed initialized method are explained as follows. First, to exploit the multiple discriminative information and the similarities between the data

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points, the model (7) is employed to construct the term $(\beta_1 \mathbf{L}_0 \mathbf{L}_0^T / \beta_2 \mathbf{L}_0 \mathbf{L}_0^T)$, with the purpose of overcoming the outlier, the half-quadratic minimization technique [14], [31] is employed to calculate the scale matrix $\mathbf{S}_1$, which is imposed on the term $(\beta_1 \mathbf{L}_0 \mathbf{L}_0^T / \beta_2 \mathbf{L}_0 \mathbf{L}_0^T)$. Second, considering the fact that EV varies spatially and spectrally, which seems as data-dependent noise. A scheme for the weighted LASSO estimators is utilized. Based on the weighted LASSO estimators [32], the Dantzig chooser is usually employed to provide a controller for the data-dependent problem. According to the Dantzig chooser, the standard deviations of the HSI matrix $\mathbf{X}$ is calculated for controlling the data-dependent noise, the data-dependent property of the Dantzig chooser is employed to supply more cues for the BHU, hence the term $((\beta_1 \mathbf{L}_0 . * \mathbf{S}_2 \mathbf{L}_0^T)/(\beta_1 \mathbf{L}_0 . * \mathbf{S}_2 \mathbf{L}_0^T))$ is obtained by integrating the theory of model (7) with the Dantzig chooser, and together with the term $((\beta_1 \mathbf{S}_1 . * \mathbf{L}_0 \mathbf{L}_0^T)/(\beta_2 \mathbf{L}_0 \mathbf{L}_0^T))$, $((\beta_1 \mathbf{S}_1 . * \mathbf{L}_0 \mathbf{L}_0^T + \beta_1 \mathbf{L}_0 . * \mathbf{S}_2 \mathbf{L}_0^T)/(\beta_2 \mathbf{L}_0 \mathbf{L}_0^T + \beta_1 \mathbf{L}_0 . * \mathbf{S}_2 \mathbf{L}_0^T))$ is formulated, in such a way the initializing scheme for the projection matrix $\mathbf{P}$ is expected to pursue a multidiscriminative subspace by mitigating the influence of EV. Thirdly, the nonlinearity is tackled in the proposed initializing method, where the manifold learning is employed to obtain a faithful space, that is, the term $(\alpha \mathbf{W} / \alpha \mathbf{D})$ in the expression (20) aims to control the influence of the nonlinearity.

IV. Experimental Environment

To validate the performance of the proposed unmixing method, a great number of experiments are implemented on synthetic and several broadly used real data sets,[4] which are shown in Figs. 1–6.[5] The characteristics of the tested data sets are clarified as follows.

1) To simulate the synthetic data, five reference endmembers are randomly chosen from the United States Geological Survey spectral library. The synthetic data sets are simulated with 100 × 100 pixels and 224 spectral bands, the generated abundance maps follow Dirichlet distribution, which are designed to satisfy the constraints of nonnegativity and sum-to-one, and three challenging scenarios are simulated to ensure the absence of pure endmember pixels and with different perturbations. Fig. 1 shows false color images of the synthetic data sets, specifically, Fig. 1(a) is a clean synthetic image without any perturbation; Fig. 1(b) is corrupted by additive noise; Fig. 1(c) is corrupted by outliers and additive noise; Fig. 1(d) is corrupted by outliers, additive noise, EV, and nonlinearity, and the images in Fig. 1(b)–(d) denote three scenarios, which are tested in this article, it is visible that the images Fig. 1(c) and (d) are terribly corrupted with increasing interferences.
    a) In the first scenario, for example, Fig. 1(b), additive noise is added, and the additive noise follows zero-mean white Gaussian distribution, and SNR can be calculated as follows:
    $$\mathrm{SNR} = 10 \log_{10} \frac{\mathbb{E}\left[\mathbf{X}^T \mathbf{X}\right]}{\mathbb{E}\left[\mathbf{E}^T \mathbf{E}\right]}. \tag{23}$$
    $\mathbb{E}$ is an operator to calculate the mathematical expectation.
    b) In the second scenario, shown in Fig. 1(c), additive noise and outliers are added. To simulate the outliers, SNR of the additive noise is fixed to 25, 10%∼35% of the pixels of each synthetic image by randomly saturating 67 bands, where the reflectance values are replaced by 0.
    c) In the third scenario, additive noise, outliers, EV, and nonlinearity are considered, shown in Fig. 1(d). To simulate these multiple interferences, spectral signatures are multiplied via randomly generated scaling factors, which are in the range of [0.6, 1.3]. Then nonlinearity is simulated by a blurring kernel, which is inspired by the work of Wang *et al.* [33]. The blurring matrix $\mathbf{Z} \in \Re^{B \times B}$ is generated by 4×4 Gaussian point spread function with variances 0.2, and a bilinear mixing model [33] is taken, that is, $\mathbf{X} = \mathbf{ZUV}$. After that, 30% outliers and additive noise (SNR = 25) are added.
2) Unlike the synthetic data sets whose reference endmember signatures and ground truth are available, the real ground truths of the real HSI sets are absent. To test the performances of the compared methods, we download the reference ground truth for the Jasper and Urban data sets at.[4] The involved real HSI sets are introduced as follows.
    a) The first one is shown in Fig. 2(a), which has been used in [12]. It is acquired in June 2012, and it covers the University of Houston campus located in TX, USA.[6] The subset contains 144 spectral bands in the range of 380–1050 nm, and the subimage contains 105 × 128 pixels, there are four endmembers in the subimage, that is, vegetation, red roofs, concrete, and asphalt.

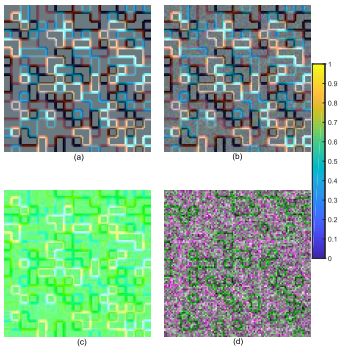


Fig. 1. Pseudocolor images of the synthetic data sets.[6] (a) Clean image without any interference. (b) Additive noise is added to (a), and SNR = 25. (c) Outliers are added to (b), and the percentage of the outliers is 30%. (d) Interferences of EV and nonlinearity are added to (c).

[4]http://lesun.weebly.com/hyperspectral-data set.html

[5]The bands employed in the false color figures as RGB channel are channels (58, 33, 21).

[6]http://hyperspectral.ee.uh.edu/

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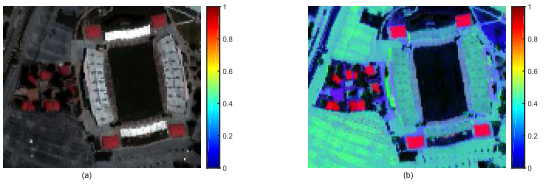


Fig. 2. (a) Pseudocolor images of Houston data sets,[6] and (b) corresponding latent subspace image used in the experiments.

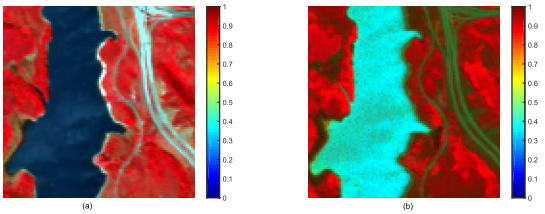


Fig. 3. (a) Pseudocolor images of Jasper with 224 bands and (b) corresponding latent subspace image used in the experiments.

b) The second one is Jasper data set, which is shown in Fig. 3(a). In the experiments, the subimage contains 100 × 100 pixels, and each pixel is collected at 224 channels in the range of 380–2500 nm. There are four endmember signatures in the data set, that is, tree, water, dirt, and road. Usually, before unmixing, the corrupted channels are removed. In our experiments, to demonstrate the robustness of the proposed method, the data set is employed without any preprocessing, and a subimage 100 × 100 with 224 bands is used.
c) The third one in Fig. 4(c) and the fourth one in Fig. 5(a) are both Urban data sets. Urban is one of the most widely utilized HSIs in the BHU study. There are 307 × 307 pixels in the subimage across 210 wavelengths varying from 400 to 2500 nm, the spatial and spectral resolutions are 2 m and 10 nm, respectively. In particular, the difference between Figs. 4(c) and 5(a) is the number of bands, the former is preprocessed by removing the corrupted bands, that is, the channels 1∼4, 76, 87, 101∼111, 136∼153, and 198∼210 are removed, that is, 162 channels are remained to unmix. To demonstrate the methods of unmixing the degenerated data set, the original Urban data set with 210 bands is also employed in this article, the pseudocolor image of which is shown in Fig. 5(a). According to the reference ground truth,[4] there are six endmembers in the data set, respectively, are asphalt, grass, tree, roof, metal, and dirt.
d) The last one is the most popular data sets for the BHU research that is located in the Cuprite in Las Vegas, NV, USA [see Fig. 6(a)]. The original data set contains 224 channels, varying from 370 to 2480 nm. Generally, before unmixing, the noisy channels and water absorption channels are removed. In the experiments, a subset of 188 channels with 250 × 191 is considered. According to the existing literature [14], [34] and,[7] the number of endmembers is considered as 12.

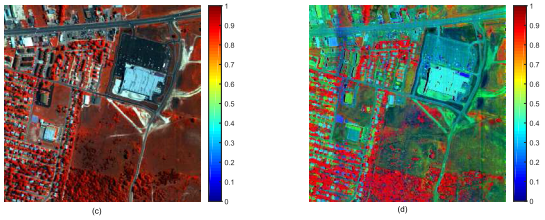


Fig. 4. (a) Pseudocolor images of the preprocessed Urban data sets with remaining 162 bands and (b) corresponding latent space image.

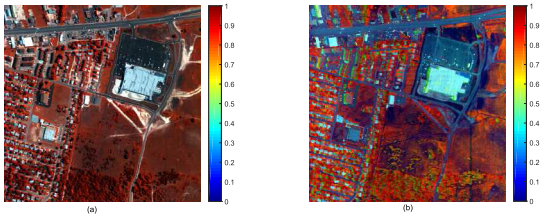


Fig. 5. (a) Pseudocolor images of the gross Urban data sets with 210 bands and (b) corresponding latent subspace image.

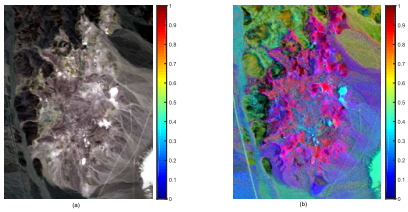


Fig. 6. (a) Pseudocolor images of Cuprite data sets and (b) corresponding latent space image used in the experiments.

To illustrate the effectiveness of the constructed latent space, we show all the latent spaces of the real data sets (see Figs. 2–6), specifically, in each group figures, the left one is the original HSI, and the right one is the constructed latent space that corresponds to the left original image. In contrast to the original ones, the merits of the reconstructed underlying ones are twofold. ⟨1⟩ Interferences are overcome by the manifold learning, and intrinsic features are caught by the sparse constraints on the projection matrix $\mathbf{P}$ and transition matrix $\mathbf{A}$. ⟨2⟩. More discriminative subspace is obtained by the theory of the subspace clustering. According to Figs. 2–6, it is easily concluded that the constructed underlying subspace can obtain higher fidelity features, which are more discriminative than the corresponding original tested data set, thus, the superior performance of the subsequent unmixing can be guaranteed by the exploited latent subspace.

To calculate the graph weight matrices, we set the kernel parameter τ of the proposed method as 0.1, and set the number of the KNN as 10. Additionally, in all the experiments, using the initializing (20) to initialize the projection matrix $\mathbf{P}$, and $\beta_1 = 1e-3$, $\beta_2 = 1e-13$, the initial latent matrix

[7]https://speclab.cr.usgs.gov/cuprite95.tgif.2.2um_map.gif

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TABLE I
COMPARED METHODS

GLMM	Based on the extended linear mixing model [12], more complex spectral distortions are considered by the *Generalized Linear Mixing Model (GLMM)* [35]. Considering GLMM is very time consuming and can not calculate the endmember matrix directly, we just show the estimated abundance maps by GLMM on some data sets.
EA-NMF	A sparse constrained BHU method via approximating the reconstruction error, called EA-NMF. In EA-NMF, the outliers and the noises are tackled [14].
rNMF	An extended LMM-based unmixng method by employing an additional term to address the possible nonlinear effects [36].
RSU	An abundance map estimating method with $l_{2,1}$ norm for mitigating the influence of noises and outliers [37].
MCSNMF	Cauchy sparse NMF with manifold graph regularization is robust to noise [38].
RCONMF	It uses a regularization term to boost minimum volume by promoting the endmember signatures toward the average value of the data point or bring the endmember signatures to the actual solution quadratically constrained by a given simplex [39].
RCMF	A robust constrained matrix factorization approach to outliers for the BHU via addressing the association between the endmember signatures and the observed data [17] (In this paper, sparsity level in RCMF for the nonnegative subspace pursuit approach is set as $K+4$.).

$\mathbf{L}$ is set as the input HSI matrix, that is, $\mathbf{L}_0 = \mathbf{X}$. Seven existing methods are employed to compare, in these compared methods, the interference, i.e., either nonlinearity, outlier or EV is considered, some particulars of all the compared methods are listed in Table I. Except for the specified parameter in Table I, all the related parameters in the compared methods are fitted as the same values of the corresponding referred literature. In consideration of the ill-posedness of the BHU, each method is implemented repeatedly 15 runs, the means and standard deviations of the metric values are listed, and the best results are marked by bold fonts, moreover, each run of the comparison is under the circumstance of a unified initializing strategy. All the experiments are conducted via MATLAB 2018a code, on a platform of 2.9-GHz Intel Core CPU server with windows 10 and 64-GB memory.

To evaluate the proposed method quantitatively, we use two widely utilized measure metrics, that is, SAD and RMSE. The expressions of SAD and RMSE are written as follows:

$$\mathrm{SAD}_k = \cos^{-1}\left(\frac{u_k^T \widehat{u}_k}{\|u_k\|\|\widehat{u}_k\|}\right) \tag{24}$$

$$\mathrm{RMSE}_k = \left(\frac{1}{N}|v_k - \widehat{v}_k|^2\right)^{\frac{1}{2}} \tag{25}$$

where $\widehat{u}_k$ is the actual endmember and its estimate u_k, $\widehat{v}_k$ and v_k are, respectively, the kth real abundances and estimated abundances.

V. RESULTS AND ANALYSES

In this section, results on the aforementioned synthetic and real data sets are illustrated and the corresponding analyses are given. Note that all the tested unmixing algorithms are initialized by the following strategy, that is, the endmember matrix is initialized by the way of randomly initialization, and the corresponding abundance matrix is initialized by SCLSU [30].

A. On Synthetic Set

This section tests the performance of the advocated unmixing algorithm when considering one or two or multiple perturbations. AVG − SAD and AVG − RMSE denote average values of SAD and RMSE.

1) Third Scenario: In this scenario, the influence of the tradeoff parameter α is tested on multiple perturbations corrupted data sets, and the performance of the proposed method is investigated with the varied number of the iterations, to avoid the influence of the initializations, all the tests are conducted with the same initialization. Based on the analysis on the parameter and the varied number of the iterations, the robustness to the multiple perturbations is researched. In particular, the perturbations include additive noise, outliers, EV, and nonlinearity.

To comprehensively investigate the effects of the related constraints, the performances of the constraints $(1/2)\mathrm{Tr}(\mathbf{P}^T\mathbf{C}\mathbf{P})$, $\|\mathbf{P}\|_{1/2}$, $\|\mathbf{Y}\|_{1/2}$, $\|\mathbf{V}\|_{1/2}$, and $(1/2)\mathrm{Tr}(\mathbf{V}\widehat{\mathbf{C}}\mathbf{V}^T)$ are explored separately, the unified tradeoff parameter α is replaced by the separated parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$, let them control the influences of $(1/2)\mathrm{Tr}(\mathbf{P}^T\mathbf{C}\mathbf{P})$, $\|\mathbf{P}\|_{1/2}$, $\|\mathbf{Y}\|_{1/2}$, $\|\mathbf{V}\|_{1/2}$, and $(1/2)\mathrm{Tr}(\mathbf{V}\widehat{\mathbf{C}}\mathbf{V}^T)$, respectively. Note that α_2 directly controls the sparsity and multiheterogeneity of the latent subspace, α_3 and α_4, respectively, controls the quality of the abundance maps indirectly and directly, and the effect of α_4 is bigger than α_3. Generally, the smaller values tend to provide smooth solution, and similar contents are unable to discriminate, while larger values of α_2, α_3, and α_4 give a less noisy solution, but some local patches of the abundance map may be missed. Thus, the selection of $\alpha_2, \alpha_3, \alpha_4$ [see Fig. 7(a)] depends on the characteristic of the HSI, such as the number of the endmembers, the degree of contamination, the intensity of the mixing. α_1 and α_5 [see Fig. 7(b)] are used to maintain the nature structure of the latent space and the abundances, the larger values of them tend to give poorly separated

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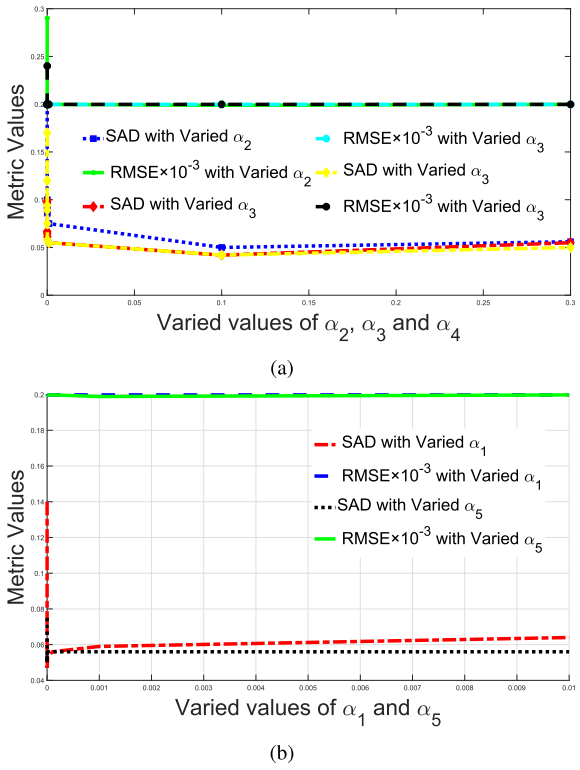


Fig. 7. Estimated results on synthetic data set with varied tradeoff parameters. (a) Metric values with varied $\alpha_2, \alpha_3, \alpha_4$. (b) Metric values with varied α_1, α_5.

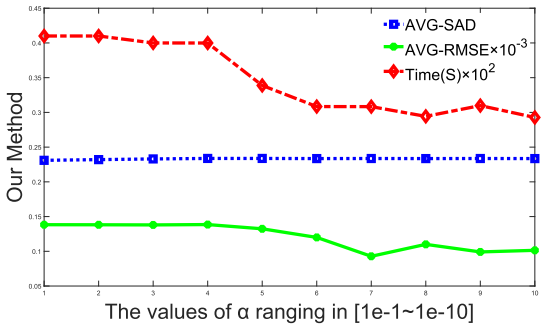


Fig. 8. Estimated metric values on the synthetic data set with varied number of iterations.

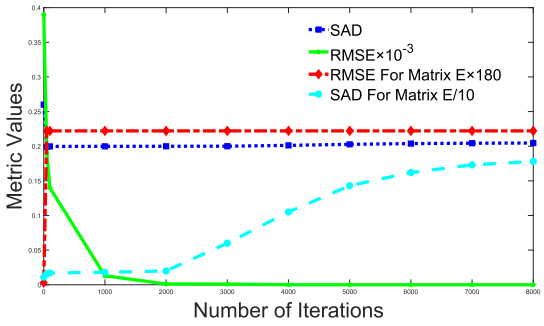


Fig. 9. Estimated metric values for the synthetic data set with varied number of iterations.

solution, and the selecting values of them depend on the level of the spatial/spectral correlation. In Fig. 7, a wide range of the values for $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, and α_5 are selected, and we can conclude that the values of SAD are insensitive to the varied parameters, but there are slight changes in the values of RMSE. Specifically, the effect of α_2 is bigger than α_3 and α_4, and α_1 is more powerful than α_5. Since the metric values almost stay stable with the varied parameters, let the tradeoff parameter of all the constraints be α for simplicity, and we set α to $1e-1 \sim 1e-10$. The metric values and convergence time are calculated, which are shown in Fig. 8. Among the results, the values of SAD are stable with the varied values of parameter α, and there are a few fluctuations in the curve of RMSE, when $\alpha = 1e-7$, the result of RMSE is best. According to the experiments with the varied parameter and the theoretical analysis, the tradeoff parameter is fine-tuned in the light of the characteristics of the HSI. Additionally, Fig. 8 also indicates that the varied parameter α can influence the convergence time, with the reducing of parameter α, the convergence time tends to decrease.

To validate the performance with the varied number of iterations, we show Fig. 9. In Fig. 9, except metric values of endmember and abundances are given, the reconstructed error matrix $\mathbf{E}$ is evaluated by the values of SAD and RMSE, which are calculated between the reconstructed error matrix $\mathbf{E}$ and HSI matrix $\mathbf{X}$. We can visually conclude that the values of SAD for endmembers and the values of RMSE for matrix $\mathbf{E}$ are stable with the increasing number of the iterations, but the values of RMSE for abundances are equal to 0 with large number of iterations. From the perspective of the denoising, when the number of iterations is 0, the values of SAD and RMSE for matrix $\mathbf{E}$ are very small, but with the increasing iterations, the values of SAD and RMSE for matrix $\mathbf{E}$ become bigger, this verifies that the proposed method can drive the noise into the reconstructed matrix $\mathbf{E}$. Note that the SAD for matrix $\mathbf{E}$ becomes bigger with the increasing number of the iterations and keep stable after 5000 iterations, that is, more noises would be driven into the matrix $\mathbf{E}$ with the increasing iterations, especially, the scale factor noise. Until all the noises are driven into the matrix $\mathbf{E}$, the SAD value for $\mathbf{E}$ will remain changeless with the increasing iterations. The phenomena indicate that the procedure can be convergent and offer better abundance maps by the sufficient iterations. Considering the time cost, in all the subsequent experiments, the number of iterations is set to 100.

Given parameter α as $1e-3$, the abundance maps and the SAD values are shown in Fig. 10 and Table II. Fig. 10 indicates that our method can estimate the abundance maps well, which almost coincide with the truth. Except for our methods, RCMF recover the abundance maps well, however, compared to the truth, there are many noises. According to Table II, the SAD values of our method is also superior than the other compared methods. Since our advocated method can capture the interferences-insensitive latent space, moreover, the proposed method can employ inherent self-expressiveness features and preserve the iteratively reconstructed collaborative similarity structure, the unmixing results in Fig. 10 and Table II prove the abovementioned advantages are exploited quite adequately on the terribly corrupted data set, and as a result, our proposed method can work well on the badly contaminated HSI.

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TABLE II

SAD VALUES BETWEEN THE GROUND TRUTH AND ESTIMATED ENDMEMBERS ON SYNTHETIC DATA SET FOR THE THIRD SCENARIO

	Our	rNMF	EA-NMF	RCONMF	$MCSNMF$	RCMF
SAD	**0.201±0.012**	0.253 ± 0.023	0.287 ± 0.025	0.255 ± 0.031	0.295±0.044	0.202±0.036

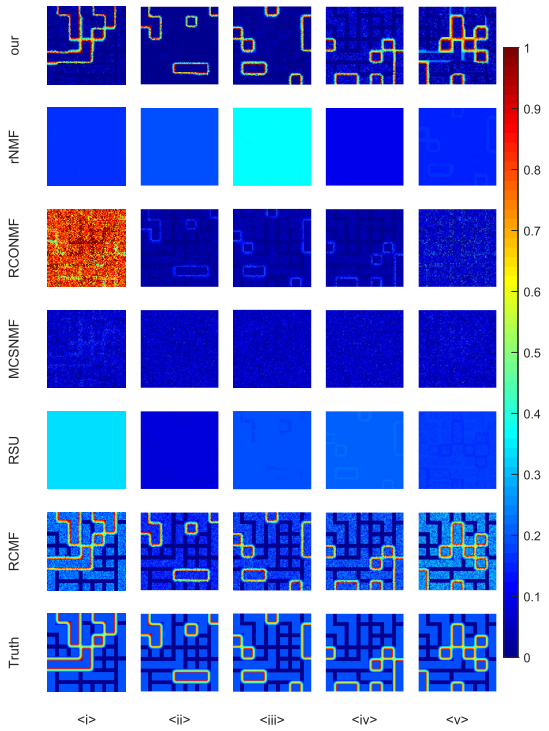


Fig. 10. Estimated and ground truth abundance maps for the synthetic data set. The color scale is 0–1(blue–red).[8]

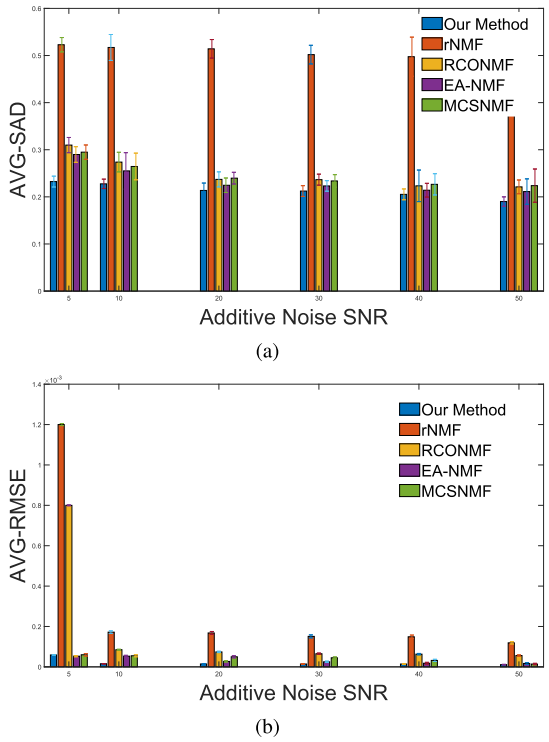


Fig. 11. Estimated results on synthetic data set with varied additive noise. (a) AVG-SAD with varied SNR. (b)AVG-RMSE with varied SNR.

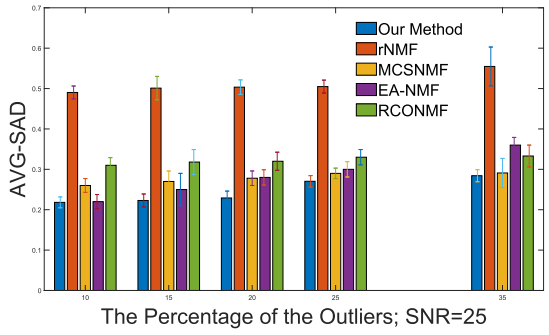


Fig. 12. AVG-SAD with varied percentages of the outliers and fixed SNR.

For the subsequent experiments on the synthetic HSIs, according to the level of the contamination, α is fixed as $1e-4$. In our proposed method, in the second and third scenarios, the matrix Y of our method is initialized as $(V^T/2)$.

2) First Scenario: Five unmixing methods are employed to evaluate the performance on additive noise corrupted data sets, the results are shown in Fig. 11. In Fig. 11, SNR ranges from 5 to 50. It is clear that the average values of SAD and RMSE for all the compared methods decrease with the increasing values of SNR, especially, our method performs best in all the compared methods. It demonstrates that the additive noise can deteriorate the unmixing results and our advocated method is robust to the additive noise.

3) Second Scenario: Here, additive noise is added, and SNR is fixed as 25, besides, outliers are added. To evaluate the robustness of the proposed method on two perturbations corrupted data sets, the percentage of the outliers are set to 10%, 15%, 20%, 25%, and 35%. The values of SAD are shown in Fig. 12. Although our proposed method can obtain the best results in all the compared methods, it becomes worse when the percentage of the outliers is bigger.

B. On Real Data Sets

In this section, the experiments are conducted on Houston, Jasper, Urban, and Cuprite data sets. For each data set, the approximation error of the proposed method almost keeps the same after a number of iterations, that is, the proposed method is experimentally proven to be convergent.

1) On Houston Data Set: The tradeoff parameter is set as $\alpha = 1e-10$. Specifically, to test performance with different initialized matrix Y, on Houston data set, matrix Y is initialized by two ways, that is, randomly initialization and $(V^T/2)$. Based on the unified initializing scheme, the matrix Y in our proposed method is set the way of randomly initialization, the initial components of Y are assigned between 0 and 1 randomly. The estimated abundance maps by the tested

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TABLE III

MEAN TIME COST OF EACH RUN AND SAD VALUES BETWEEN THE GROUND TRUTH[7] AND ESTIMATED ENDMEMBERS ON HOUSTON DATA SET

	Our	EA-NMF	RCONMF	MCSNMF	RCMF
vegetation	0.0264±0.002	0.0435 ± 0.004	0.0536 ± 0.006	0.0078±0.002	**0.0078±0.001**
red roofs	**0.0309±0.002**	0.0479 ± 0.007	0.0395 ± 0.006	0.068 ± 0.007	0.0364 ± 0.005
concrete	**0.0025±0**	0.0386 ± 0.002	0.0661 ± 0.004	0.07 ± 0.006	0.067 ± 0.005
asphalt	0.0601±0.004	0.0764 ± 0.005	0.0997 ± 0.007	0.095 ± 0.008	**0.0505±0.003**
$\overline{SAD}$	**0.0356±0.002**	0.0516 ± 0.005	0.0647 ± 0.006	0.06 ± 0.006	0.0404 ± 0.004
Time Cost(S)	15.91±0.95	23.43 ± 0.48	**5.47±0.29**	66.89 ± 6.55	42.55 ± 3.41

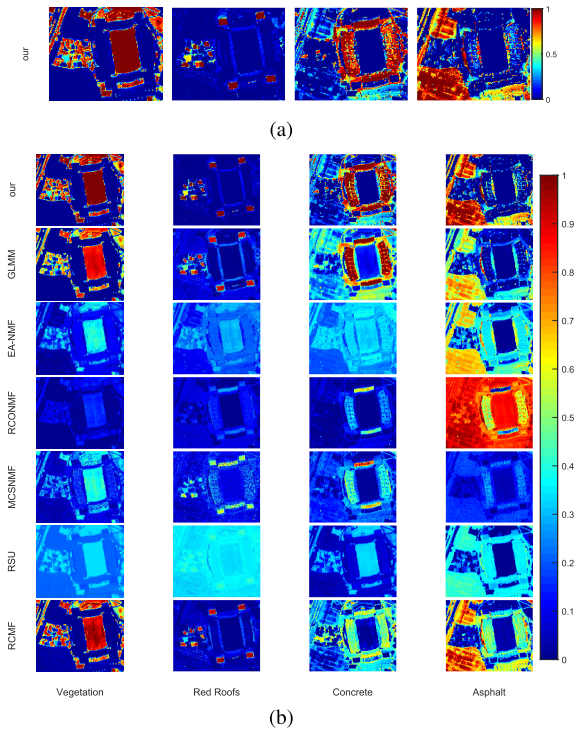


Fig. 13. (a) Estimated abundances by our method with $Y = V^T/2$. (b) Estimated abundances by several methods.[8]

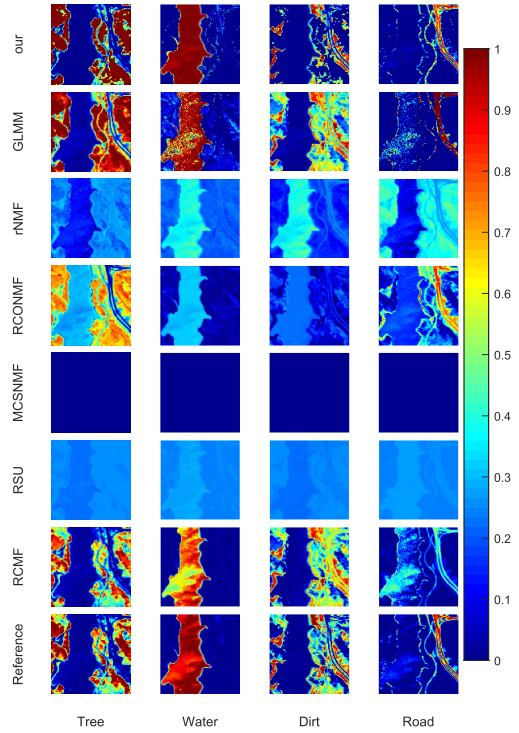


Fig. 14. Reference ground truth[4] and the extracted abundance maps for the Jasper data set. The color scale ranges from 0 to 1(blue to red).[8]

methods are shown in Fig. 13(b),[8] and the values of SAD and time cost are listed in Table III. It can be concluded from Fig. 13(b), that the estimated abundance maps via our proposed method well coincide with the ground truth [12]. It is clear that the other methods fail to extract the abundance maps of concrete and asphalt, as a whole, the estimated abundance maps by our proposed method are better than the estimated ones by the other methods.

Since the ground truth of the abundances are absent, just the mean values and standard deviations of SAD values and the time costs for the tested methods are listed in Table III, where $\overline{\text{SAD}}$ is the mean value of the endmember SADs. Table III shows that our proposed method identifies most of the endmembers, and the average accuracy of the identification is too much superior than the other methods. However, our method is not the most time-saving method for the higher computational complexity.

Moreover, we give the abundance results with the same initializations as above endmembers and abundances, but initialize matrix Y by $(V^T/2)$, and show the results in Fig. 13(a). We can visually conclude that the results in Fig. 13(a) are slightly purer than the results in the first line of Fig. 13(b), which are obtained by our proposed method with randomly initialized matrix Y. The possible reasons are explicated as follows. The proposed method exploits the theory of subspace clustering, the matrix YV is equal to the discriminative matrix A, which records the relationship among the pixels. Moreover, Y collects the relationship between the endmembers and the pixels, and it is a sparse matrix. In this sense, matrix Y and matrix V are closely related. Since the abundance matrix V is initialized by an effective sparse method, that is, SCLSU. Hence, initializing matrix Y by $(V^T/2)$ can obtain a sparse matrix, in contrast to the randomly initialization, random errors can be mitigated, as a result, $(V^T/2)$ is a bit better for initializing matrix Y. Therefore, the experiments on Jasper and Urban data sets, the matrix Y is initialized as $(V^T/2)$.

2) On Jasper Data Set: For our proposed method, the parameter is set to $\alpha = 1e-4$. The estimated abundance maps of all the selected methods and the reference ground truth[4] are shown in Fig. 14, moreover, the SAD values (the SAD values for MCSNMF and EA-NMF are nonnumerical, we just show the SAD values of four methods) and the time cost of each method are listed in Table IV. Although there are

[8]A red pixel denotes the contribution of the substance is large; Conversely, a blue pixel denotes the small contributed substance. Therefore, pure red represents that the proportion of a certain kind of substance in this pixel is 1, while pure blue represents 0.

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TABLE IV

TIME COST AND SAD VALUES BETWEEN ESTIMATED ENDMEMBERS AND THE REFERENCE GROUND TRUTH[4] ON JASPER DATA SET

	Our	rNMF	RCONMF	RCMF
tree	**0.0337±0.004**	0.2474 ± 0.038	0.2156 ± 0.019	0.0516 ± 0.007
water	**0.0973±0.011**	0.2915 ± 0.024	0.3961 ± 0.042	0.1059 ± 0.072
dirt	**0.0298±0.003**	0.1986 ± 0.024	0.8247 ± 0.061	0.061 ± 0.003
road	0.1128±0.011	0.1713 ± 0.025	0.1672 ± 0.033	0.3414 ± 0.042
$\overline{SAD}$	**0.0684±0.007**	0.2272 ± 0.028	0.4009 ± 0.039	0.14 ± 0.031
Time Cost(S)	22.3±2.1	**0.281±0.02**	6.56±0.7	43.22 ± 0.57

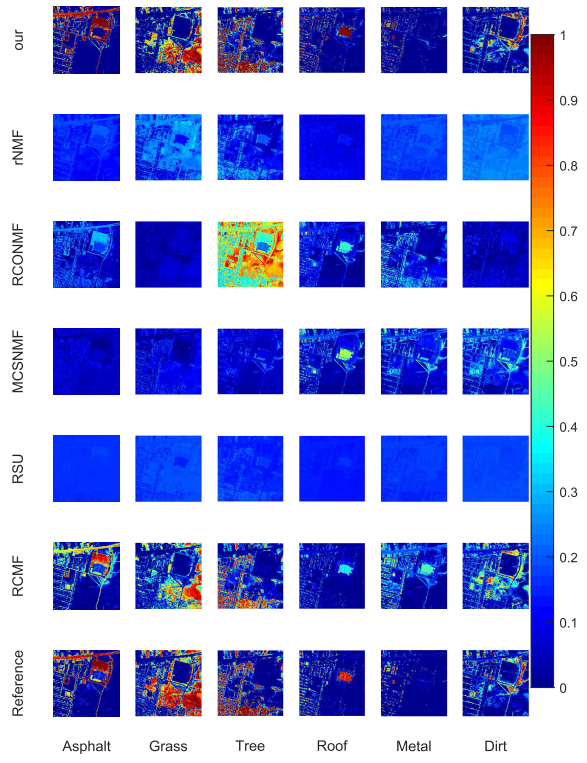


Fig. 15. Estimated abundance maps and the reference ground truth[4] for the 162 bands contained Urban data set. The color scale ranges from 0 to 1(blue to red).[8]

contaminated bands in the Jasper data set, which is not preprocessed before unmixing, Fig. 14 apparently shows that the estimated abundance maps by our method well coincide with the reference ground truth.[4] From the perspectives of the purity and accuracy, our method performs far beyond the other tested methods. Additionally, Table IV indicates that our method can extract most endmembers, and the overall accuracy of the endmember identification of our method is much higher than the other methods.

3) On Urban Data Set: In this part, the original data set with 210 bands and the preprocessed data set with 162 bands are utilized. For our method, the parameter is fixed to $\alpha = 1e{-}4$. The obtained abundance maps and the reference ground truth are shown in Fig. 15, and the metric values and the time cost on the 162 bands and 210 bands are listed in Tables V and VI, respectively. Fig. 15 shows that the obtained abundance maps by our method are almost alike to the reference ground truth, but the abundance maps obtained by the other methods are inferior to ours. Moreover, Tables V and VI also indicate that our method on the 162 bands data set and the 210 bands data set can always obtain better values of SAD than the other methods. Surprisingly, although there are badly noisy bands in the 210 bands data set, our method and some compared algorithms even perform better than on the noisy bands removed Urban data set, such as, EA-NMF, RCONMF, and RCMF can get better values of SAD than on the 162 bands data set. On the whole, the results in Tables V and VI indicate that our method performs better on both the identification of the endmembers and the estimation of the abundance maps, hence, we can conclude that our proposed method well tackles the interferences and noises. It means that our method can quite adequately exploit the useful information to unmix, thus, the proposed method is particularly suitable for the corrupted and highly mixed data set.

4) On Cuprite Data Set: Considering the ground truth is absent, and most of the compared methods are sensitive to the initializations, we just show the abundance maps obtained by our method and RCMF. For our method, the matrix Y is initialized by the way of random initialization, and set $\alpha_1 = 1e-6$, $\alpha_2 = 1e-4$, $\alpha_3 = 1e-9$, $\alpha_4 = 1e-4$, and $\alpha_5 = 1e-5$, the estimated abundance maps are shown in the row "Our" of Fig. 16.

Referring to the existing abundance maps,[7] the abundance maps recovered by our proposed method are better than RCMF. Although RCMF extracted most of the abundance maps, there are more impure substances in the abundance maps, where the pixel color is neither blue nor red.

To further demonstrate the influence of each constraint in (9), each run with setting one or two tradeoff parameters as zero and fixing the rest of the parameters as $1e-4$. Taking the extraction of the endmember Chalcedony as an example, the results are shown in Fig. 17. Obviously, the abundance maps in Fig. 17 decline in quality in the absence of some constraint. Concretely, Fig. 17(b) is obtained on the condition of $\alpha_2 = 0$, others fixed as $1e-4$, the resulted map does not only show the distribution of Chalcedony, but also shows the other substance, which indicates that the sparse constraint on the projection matrix $\mathbf{P}$ is an important term to pursue discriminative subspace. Let $\alpha_3 = 0$, others as $1e-4$, the result is shown in Fig. 17(c), since the sparse constraint on matrix $\mathbf{Y}$ is absent, the abundance map contains impure substance and Chalcedony is not extracted wholly, which would be caused by the emergence of the artificial endmember. Fig. 17(a) and (d) are obtained in the absence of one manifold constraint, both of these two cases are incapable of wholly extracting the substance of Chalcedony, which may be caused by the interferences. However, if both the manifold constraints are absent, the purity of the map is poor, which is shown in Fig. 17(d). It shows that some non-Chalcedony emerges in the Chalcedony map. Moreover, if $\alpha_4 = 0$, Chalcedony map is unable to estimate, which is absent in Fig. 17. From Fig. 17 and above analysis, it can conclude that the influence of each constraint is unique, and the influence of sparse constraint on the matrix $\mathbf{P}$ is more crucial than the others, which indirectly indicates the necessity of exploiting the latent discriminative subspace. Additionally, the absence of the sparse constraint on the matrix $\mathbf{V}$ leads to incapability of estimating some abundance maps.

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TABLE V

TIME COST AND SAD VALUES BETWEEN THE REFERENCE GROUND TRUTH[5] AND ESTIMATED RESULTS ON URBAN WITH 162 BANDS DATA SET

	Our	rNMF	EA-NMF	RCONMF	MCSNMF	RCMF
$\overline{SAD}$	**0.143±0.016**	0.205 ± 0.034	0.229±0.018	0.407 ± 0.052	0.18 ± 0.034	0.164 ± 0.022
Time Cost(S)	198±46.1	**1.66±0.36**	43.7 ± 3.3	70.42±4.3	4467±32	495.13±67.3

TABLE VI

TIME COST AND SAD VALUES BETWEEN THE REFERENCE GROUND TRUTH[5] AND ESTIMATED RESULTS ON URBAN WITH 210 BANDS DATA SET

	Our	rNMF	EA-NMF	RCONMF	MCSNMF	RCMF
$\overline{SAD}$	**0.105±0.021**	0.208 ± 0.025	0.155±0.022	0.377 ± 0.039	0.199 ± 0.016	0.159 ± 0.026
Time Cost(S)	345±102.1	**4.1±0.6**	40.8 ± 2.3	79.49±6.7	4556.4±311.1	556.4±123.1

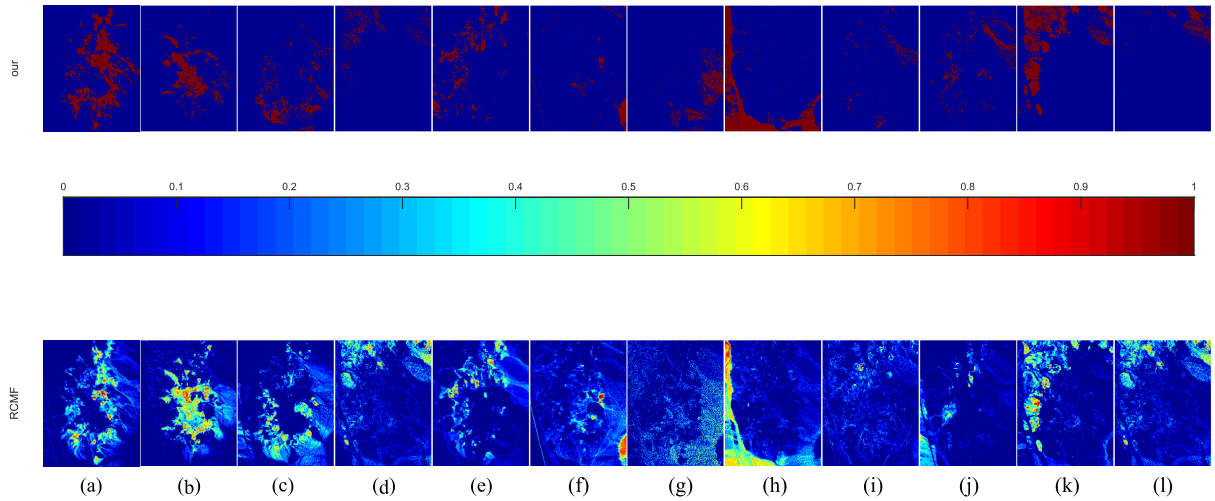


Fig. 16. Estimations of abundance maps.[8] (from top to bottom) Obtained by our method and RCMF, respectively. (a)–(l) Correspond to the abundances of 12 minerals.

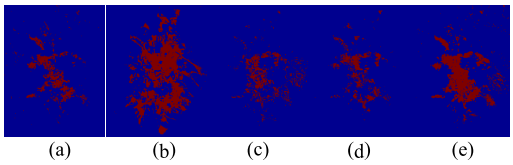


Fig. 17. (a)–(e) Estimated abundance maps for Chalcedony with varied tradeoff constraint parameters. The color scale ranges from 0 to 1(blue to red).[8]

VI. CONCLUSION

To sidestep the challenging interferences in the BHU problem, we propose a novel model to exploit underlying space from the original HSI matrix, the theory of subspace clustering is employed to develop the synergy of the similar and discriminative structures in the latent space, and the inherent self-expressiveness property is utilized to avoid the emergence of the artificial endmember spectra. Based on the proposed model, intrinsic information is acquired and sparser structure is guaranteed by the sparse constrains from multiple angles. Experimental results on synthetic and several representative real HSI sets show the effectiveness and superiority of our advocated method. Regretfully, the computational complexity of our advocated method is a little high for the two graph regularized terms and the iterations for the multiple variables, in future, we will focus on developing efficient computing.

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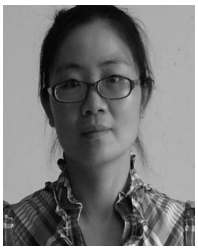
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